

Case-Study Report

Title: Incident Response and Digital Forensics Analysis

USN:

Name of Student:

Date:

Selected Case Study: [Name of Incident / Organization]

1. Case Study Overview

Objective:

To analyze a cybersecurity incident using a structured Incident Response Plan and Digital Forensics Process.

Incident Summary:

- Type of incident: [e.g., ransomware attack]
- Target environment: [enterprise network, cloud system, etc.]
- Impact: [data loss, service disruption, financial damage]

2. Incident Response Plan

2.1 Purpose

//Provide a structured method to detect, contain, eradicate, and recover from cybersecurity incidents.

2.2 Incident Response Lifecycle *//explain the following steps involved in process*

1. Preparation
2. Detection & Analysis
3. Containment
4. Eradication
5. Recovery
6. Post-Incident Review

2.3 Screenshot Requirements

//Include visuals that demonstrate the response workflow.

Screenshot 1 — Incident Response Workflow Diagram

- Flowchart showing the six IR phases
- Arrows showing decision points
- Label stakeholders at each stage

Screenshot 2 — Incident Classification Matrix

- Table showing severity levels
- Criteria: impact vs urgency

Screenshot 3 — Communication Plan

- Diagram showing reporting hierarchy
- Internal vs external notification channels

2.4 Roles in Incident Response Report

//Explain the following roles for the selected case-study

Role	Responsibility
Incident Response Manager	Leads response strategy and decision making
Security Analyst	Detects and investigates the incident
Forensic Specialist	Collects and preserves digital evidence
IT Operations Lead	Implements containment and recovery actions
Legal Advisor	Ensures regulatory and legal compliance
Communications Officer	Handles internal and external communication
Documentation Officer	Records actions, timelines, and findings

3. Digital Forensics Process

3.1 Purpose

To identify, preserve, analyze, and present digital evidence in a legally admissible manner.

3.2 Digital Forensics Lifecycle (Chain of Custody)

//explain the following steps involved in process

1. Identification
2. Preservation
3. Collection
4. Examination
5. Analysis
6. Reporting
7. Presentation

3.3 Screenshot Requirements

Screenshot 4 — Evidence Acquisition Process

- Diagram showing chain of custody
- Source → Imaging → Storage

Screenshot 5 — Forensic Investigation Workflow

- Step-by-step process from evidence identification to reporting

Screenshot 6 — Chain of Custody Form Example

- Evidence ID
- Handler names
- Time stamps
- Storage location

3.4 Roles in Digital Forensics Report

//Explain the following roles for the selected case-study

Role	Responsibility
Lead Forensic Investigator	Oversees forensic investigation
Evidence Custodian	Maintains chain of custody
Malware Analyst	Examines malicious files or code
System Examiner	Analyzes affected systems and logs
Legal Compliance Officer	Ensures evidence admissibility
Reporting Specialist	Prepares forensic report
Quality Assurance Reviewer	Validates investigation integrity

4. Roles and Responsibilities

//List all IR and forensic roles with duties.

5. Conclusion

//Impact, response effectiveness, recommendations.

References:

//List the articles, webpages, tools, books referred.